

THE UNIVERSITY OF
MELBOURNE

FNCE 30007
Derivative Securities

Lecture 10 – Interest Rate Contracts & Swaps

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Lecture Outline

- **Types of Interest Rates**
- Bond Pricing
- Forward Rates
- Forward Rate Agreement
- Interest Rate Swaps
- Valuation of Interest Rate Swaps

- *Readings: Chapter 4.5 – 4.9; Chapter 7 (currency swaps are not covered)*

Types of Interest Rates

LIBOR:

- London Interbank Offered Rate.
- A LIBOR quote by a particular bank is the rate of interest at which the bank is prepared to make a large wholesale deposit with other banks.
- One-month, three-month, six-month and twelve-month LIBOR's are quoted by large banks and financial institutions in all major currencies.
- LIBOR was often used as a proxy for the floating interest rate.
- Banks were instructed to stop writing contracts using LIBOR by the end of 2021 and all contracts using LIBOR should wrap up by June 30, 2023.

Types of Interest Rates

SOFR (US):

- Secured Overnight Financing Rate
- It replaced LIBOR following concerns about LIBOR's reliability and manipulation.
- It is a benchmark interest rate used in U.S. financial markets to measure the cost of borrowing cash overnight, collateralised by U.S. Treasury securities.
- It is considered a "nearly risk-free" rate because loans are backed by high-quality collateral (U.S. Treasuries).

BBSW (AUS):

- Bank Bill Swap Rate
- It reflects the average interest rate at which major Australian banks offer to lend and borrow short-term bank bills.
- It is a credit-based benchmark used in the Australian interest rate swap market.

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Bond Pricing

Zero Rates:

- The n-year zero-coupon interest rate is the rate of interest earned on an investment that starts today and lasts for n years.
- All of the interest and principal is realised at the end of the n years, and as such there are no intermediate or coupon payments.
- For example, suppose that $1 million is invested for 5 years with a five-year zero rate with continuous compounding of 7% p.a.
- If invested for 5 years, this investment will grow to:

  $1 million x $e^{0.07x5}$=$1.42 million

Bond Pricing

Most bonds that we can invest in, provide a periodic cash flow called a coupon, with the bond's principal (or face value) received at the maturity of the bond.

- The theoretical price of the bond is the present value of all of the cash flows received by the purchaser of the bond.

By way of example, suppose that a two-year Treasury bond with a principal of $1 million provides coupons at the rate of 8% per annum semi-annually, and the treasury zero rates are as follows:

Bond Pricing

Treasury Zero Rates	
Maturity (Years)	Zero rate % (Continuously Compounded)
0.5	6.0
1.0	6.2
1.5	6.3
2.0	6.2

Bond Pricing

What is the theoretical price of the bond:

- The bond pays coupons of:
  - $1m x 0.08/2 = $40,000 twice a year.

- The price of the bond is:

  $40{,}000e^{-0.06x0.5} + 40{,}000e^{-0.062x1.0} + 40{,}000e^{-0.063x1.5} + 1{,}040{,}000e^{-0.062x2.0} = \$1{,}031{,}521.28$

- Generalised formula

$$P = \sum c * e^{-R_i t_i} + FV * e^{-R_n t_n}$$

Bond Pricing

Par yield:

- The par yield for a certain bond is the coupon rate that causes the bond price to equal its par value (or face value).
- Example

Maturity	Zero Rate
1	2%
2	2.5%
3	3%

$$PV = c \cdot e^{-0.02*1} + c \cdot e^{-0.025*2} + (c + 100) \cdot e^{-0.03*3}$$

$$c \cdot e^{-0.02} + c \cdot e^{-0.05} + (c + 100) \cdot e^{-0.09} = 100$$

$$c = 3.054\%$$

Bond Pricing

Determining Treasury Rates:

- We can calculate Treasury rates from the quoted prices of Treasury bonds.
- For Example: Imagine that the yield curve is flat at 5% p.a. for the next 1.5 years. A 2-year bond with a principal of $1 million, that pays coupons of 10% p.a. semi-annually, is currently selling for $1.1 million. What is the two-year rate?

Bond Pricing

Step 1: Work out coupon

$$\mathrm{C} = \frac{1{,}000000}{2} \times 0.10 = \$50{,}000$$

Step 2: Discounting coupons to present using the corresponding interest rates

$$50{,}000e^{-0.05\times0.5} + 50{,}000e^{-0.05\times1} + 50{,}000e^{-0.05\times1.5} + 1{,}050{,}000e^{-R\times2}$$
$$= 1{,}100{,}000$$
$$142{,}714.1411 + 1{,}050{,}000e^{-R\times2} = 1{,}100{,}000$$
$$1{,}050{,}000e^{-R\times2} = 957{,}285.8589$$

$$-R \times 2 = ln\left(\frac{957{,}285.8589}{1{,}050{,}000}\right)$$
$$R = 0.046$$

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Forward Rates

Definition: Forward rates are interest rates implied by current zero rates for future periods.

Table 4.6 — Zero Rates and Forward Rates (Continuously Compounded)

Year (n)	Zero Rate (% p.a.)	Forward Rate for nth Year (% p.a.)
1	3.0	—
2	4.0	5.0
3	4.6	5.8
4	5.0	6.2
5	5.3	6.5

General Formula

R_1 = zero rate for T_1, R_2 = zero rate for T_2

Forward rate for period $T_1 \rightarrow T_2$:

$$RF = (R_2T_2 - R_1T_1) / (T_2 - T_1)$$

Key: with continuous compounding, combining rates across successive periods gives an overall rate equal to the weighted average. Only approximately true for other compounding conventions.

Worked Examples

Year 2 Forward Rate

RF = (4.0×2 − 3.0×1)/(2−1) = (8.0−3.0)/1 = 5.0% p.a.

Verification: $100e^{0.03*1} \times e^{0.05*1} = 100e^{0.04*2} =$ \$108.33 ✓

Investing at 3% (yr 1) then 5% (yr 2) = 4% overall for 2 yrs.

Year 3 Forward Rate

RF = (4.6×3 − 4.0×2)/(3−2) = (13.8−8.0)/1 = 5.8% p.a.

Investing at 4.0% for 2 yrs then 5.8% for yr 3 gives same return as 4.6% for 3 yrs.

Check: (4.0×2 + 5.8×1)/3 = 13.8/3 = 4.6% ✓

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Forward Rate Agreements

A forward rate agreement (FRA) is an agreement that a certain rate will apply to a certain principal during a certain future time period.

- FRAs are over the counter interest rate derivatives.
- Two parties agree to guarantee each other a given interest rate at a future date.
- A notional principal is used but not exchanged.
- When the settlement date arrives one party compensates the other party.

An FRA is equivalent to an agreement where interest at a predetermined rate, R_K is exchanged for interest at the market rate.

Forward Rate Agreements

Example:

Suppose that a company enters into an FRA that is designed to ensure it will receive a fixed rate of 4% on a principal of $100 million for a three-month period starting in three years. If three-month floating proves to be 4.5% for the three-month period, the cash flow to the lender will be

100,000,000*(0.04-0.045)*0.25=-125,000

The lender needs to pay his counterparty $125,000 at the 3.25-year point.

Forward Rate Agreements

Valuation of FRA

When the FRA is designed at time zero, the value of FRA is ZERO.

The fixed rate R_K is equal to the forward floating rate R_F.

As time passes, R_K remains the same but R_F changes, the contract therefore no longer has a value of zero.

Consider an FRA where a company will receive a rate of 6% compounded annually, on a principal of $100 million between the end of year 1 and the end of year 2. The forward rate is 5% with continuous compounding or 5.127% with annual compounding.

The value of the FRA is 100,000,000 × (0.06 − 0.05127)e^−0.04 × 2 = $805,800

We will use this valuation approach to value swaps, which are treated as portfolios of FRAs.

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Interest Rate Swaps

Interest rate swaps are an agreement where a floating rate is exchanged for a fixed rate or vice versa.

For Example: you have borrowed some money at SOFR and now want to swap to a fixed rate exposure.

Interest rate swaps have a number of benefits attached to them:

- They can change your exposure from floating to fixed or fixed to floating; and,
- They are able to reduce the costs of borrowing.

Interest Rate Swaps

What is it?

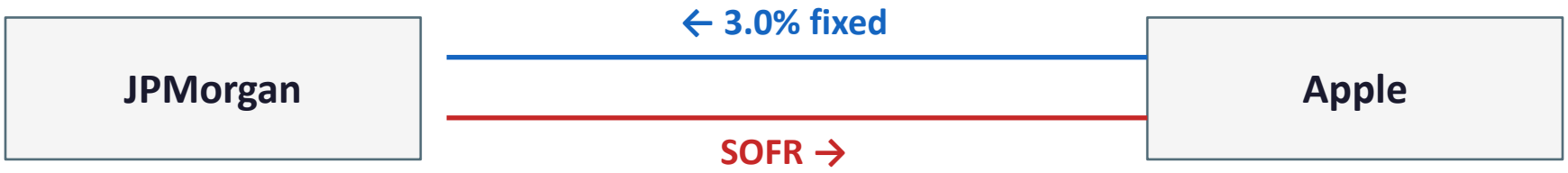
An agreement to exchange interest payments on a *notional principal* — the principal itself is never exchanged.

One party pays **fixed**; the other pays **floating (SOFR)**.

Payments are **netted** — only the difference changes hands.

Example: Apple & JPMorgan

Notional: $100 million
Term: 3 years (Mar 2023 – Mar 2026)
Payments: Every 6 months
Fixed rate: 3% p.a. (semiannual)
Floating: SOFR (compounded in arrears)



Apple = fixed-rate payer | JPMorgan = floating-rate payer

Interest Rate Swaps

Exchange 1 — September 8, 2023 (SOFR over Mar 8 – Sept 8, 2023 = 2.2%)

Apple pays (fixed)

$0.5 \times 3\% \times \$100M = \$1.5M$

→ **Apple pays $1.5 million**

JPMorgan pays (floating)

$0.5 \times 2.2\% \times \$100M = \$1.1M$

→ **JPMorgan pays $1.1 million**

Net: Apple pays JPMorgan $\$1.5M - \$1.1M =$ **$0.4 million**

Exchange 2 — March 8, 2024 (SOFR Sept 8, 2023 – Mar 8, 2024 = 2.8%)

Apple pays (fixed)

$0.5 \times 3\% \times \$100M = \$1.5M$

→ **Apple pays $1.5 million**

JPMorgan pays (floating)

$0.5 \times 2.8\% \times \$100M = \$1.4M$

→ **JPMorgan pays $1.4 million**

Net: Apple pays JPMorgan $\$1.5M - \$1.4M =$ **$0.1 million**

Interest Rate Swaps

Full Cash Flow Schedule — Apple's Perspective

Date	6M SOFR (%)	Floating Received ($M)	Fixed Paid ($M)	Net ($M)
Mar 8, 2016	2.20	—	—	—
Sept 8, 2016	2.80	+1.10	−1.50	**−0.40**
Mar 8, 2017	3.30	+1.40	−1.50	**−0.10**
Sept 8, 2017	3.50	+1.65	−1.50	**+0.15**
Mar 8, 2018	3.60	+1.75	−1.50	**+0.25**
Sept 8, 2018	3.90	+1.80	−1.50	**+0.30**
Mar 8, 2019		+1.95	−1.50	**+0.45**

Note: The $100M principal is used only for calculating interest — it is never actually exchanged.

Interest Rate Swaps

Using Swaps to Transform Liabilities & Assets

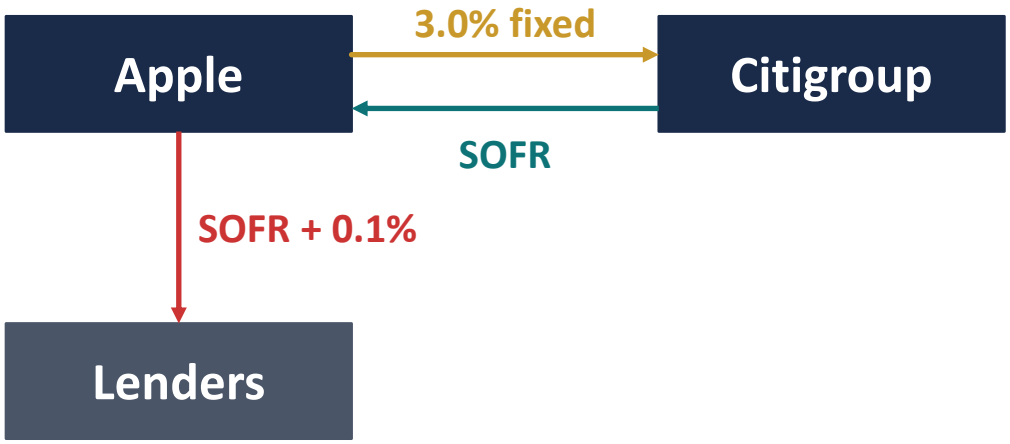
A swap can change the nature of borrowings or investments without refinancing

TRANSFORM A LIABILITY	TRANSFORM AN ASSET
Floating loan → Fixed loan	Fixed investment → Floating
Fixed loan → Floating loan	Floating investment → Fixed

Interest Rate Swaps

Transforming a Liability: Floating → Fixed

Apple borrows floating, swaps into fixed



Net cash flows:

– (SOFR + 0.1%) (pays lender)

+ SOFR (receives swap)

– 3.0% (pays swap)

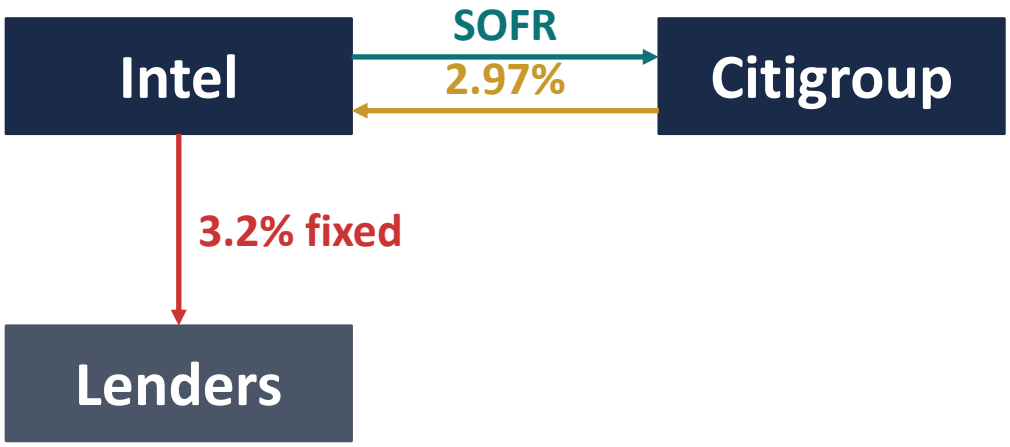
= 3.1% fixed

Before: SOFR + 0.1% → After: 3.1% fixed

Interest Rate Swaps

Transforming a Liability: Fixed → Floating

Intel borrows fixed, swaps into floating



Net cash flows:

– 3.2% (pays lender)
+ 2.97% (receives swap)
– SOFR (pays swap)

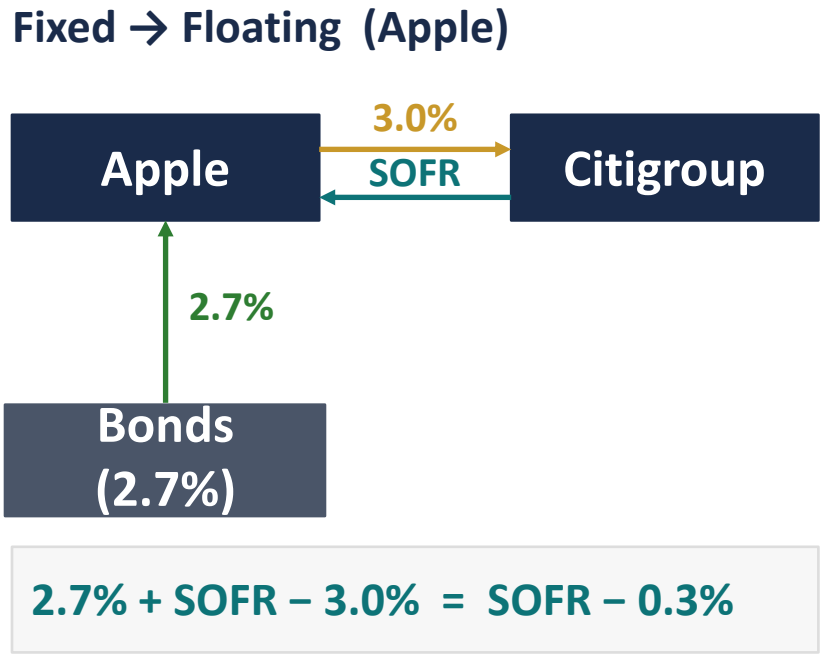
= SOFR + 0.23%

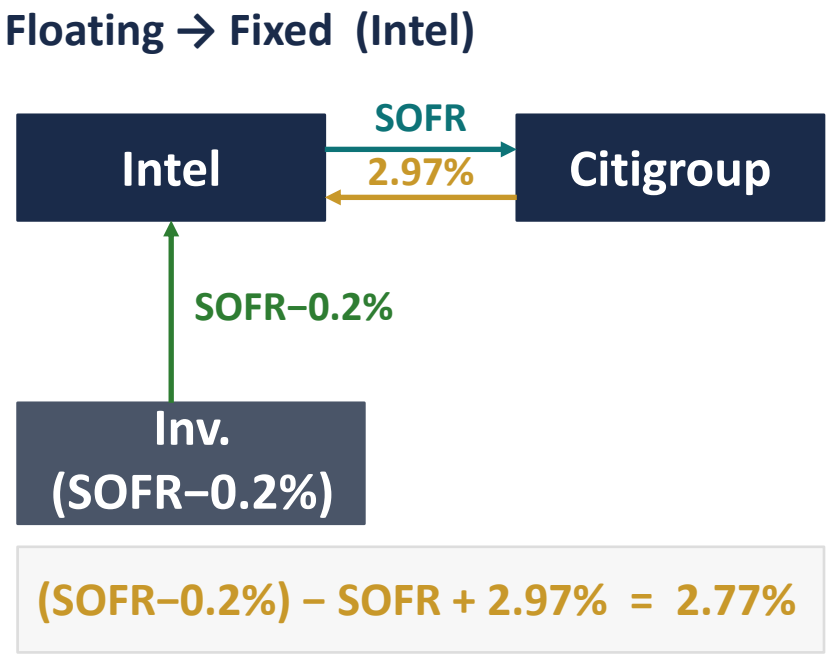
Before: 3.2% fixed → After: SOFR + 0.23%

Interest Rate Swaps

Transforming an Asset

Same mechanism — swap receipts alter the effective return





The swap receipts/payments cancel out part of the original cash flow, leaving a transformed rate

Interest Rate Swaps

By way of example, let's imagine that:

Company A can borrow:

Fixed: 7%

Floating: SOFR +1%

But really wants a floating rate

Company B can borrow:

Fixed: 10%

Floating: SOFR + 2%

But really wants a fixed rate

How can we design a swap that gets both parties what they want, and makes them better off?

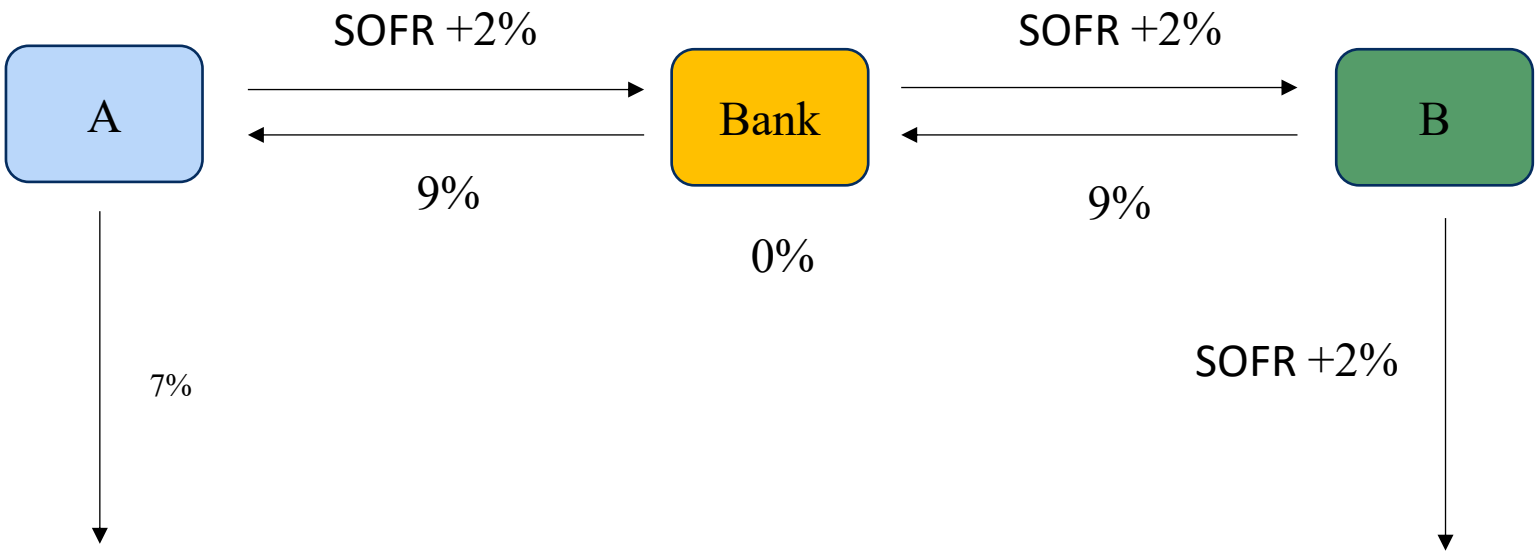
Interest Rate Swaps

- The solution to this problem is to examine where each party's comparative advantage lies:

	Fixed	Floating
Company A	7%	SOFR+1%
Company B	10%	SOFR+2%
Difference	3	1

- Maximum saving is difference in differences: 2%
  - Company A has a comparative advantage in fixed
  - Company B has a comparative advantage in floating

Interest Rate Swaps



A's borrowing cost	
Fixed	7%
Floating	SOFR + 1%

B's borrowing cost	
Fixed	10%
Floating	SOFR + 2%

Interest Rate Swaps

The cost of borrowing after entered into IRS:

Company A	Company B
-7%	-(SOFR+2%)
+9%	+(SOFR+2%)
-(SOFR+2%)	-9%
-SOFR	-9%

✓Company A is better off by 1% (i.e.,SOFR+1% − SOFR)

✓Company B is better off by 1% (i.e., 10% − 9%)

✓Therefore, the total gain is 2%

Interest Rate Swaps

What if intermediary requires 0.5%?

Company A		Company B		Difference
Fixed	7%	Fixed	10%	3%
Floating	SOFR + 1%	Floating	SOFR + 2%	1%
				2%

Given that the intermediary takes 0.5% p.a. as commission, the total gain is:

$$2\% - 0.5\% = 1.5\%$$

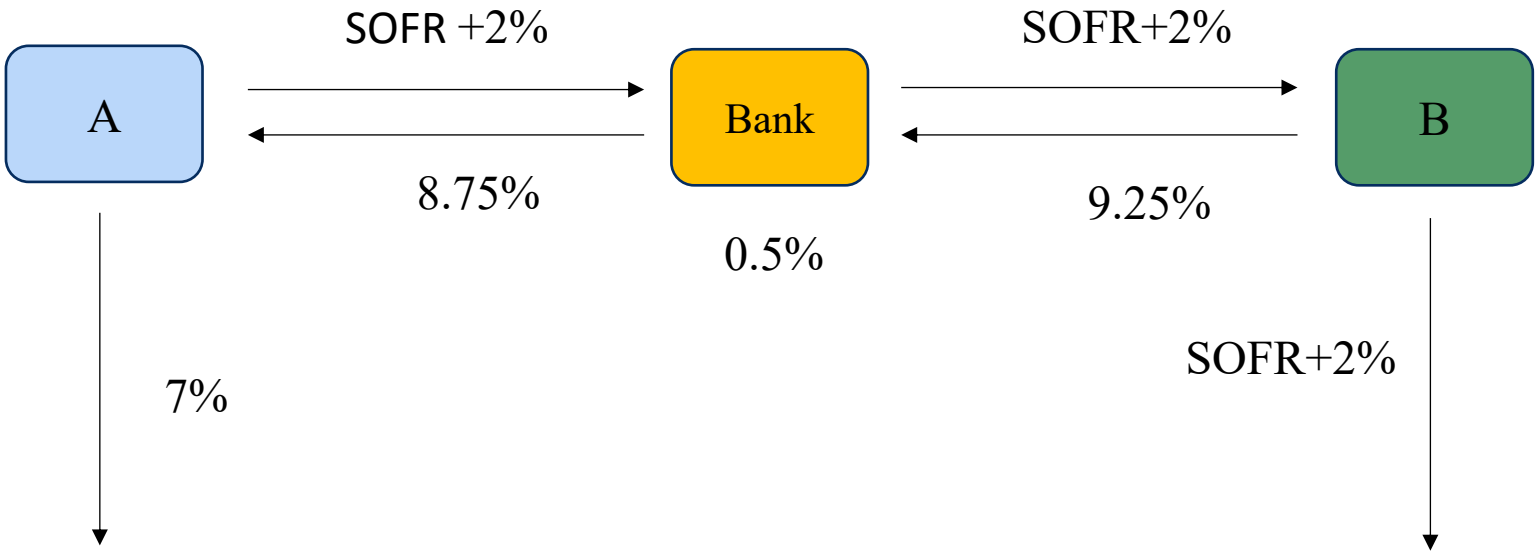
Because we need to design a swap that is equally attractive to both companies:

$$1.5\% \div 2 = 0.75\%$$

✓Company A needs floating: SOFR+0.25% (i.e.,SOFR+1% − 0.75%)

✓Company B needs fixed: 9.25% (i.e., 10% − 0.75% = 9.25%)

Interest Rate Swaps



A's borrowing cost	
Fixed	7%
Floating	SOFR + 1%

B's borrowing cost	
Fixed	10%
Floating	SOFR + 2%

Interest Rate Swaps

The cost of borrowing after entered into IRS:

Company A	Company B
-7%	-(SOFR+2%)
+8.75%	+(SOFR+2%)
-(SOFR+2%)	-9.25%
-(SOFR+0.25)%	-9.25%

✓Company A is better off by 0.75% (i.e., $\text{SOFR} + 1\% - \text{SOFR} - 0.25\%$)

✓Company B is better off by 0.75% (i.e., $10\% - 9.25\%$)

✓Therefore, the total gain is 1.5%

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Valuation of Interest Rate Swaps

Each exchange of payments is a Forward Rate Agreement (FRA)

The Key Insight

A swap = portfolio of FRAs

Each future cash exchange is equivalent to one FRA where fixed is swapped for the floating rate (or the other way around).

Value at the time of entry ≈ zero

The fixed rate is set so the swap has zero value when first negotiated.

Value changes over time

As rates move, some FRAs gain value, others lose. The swap's net value can become positive or negative.

Valuation Procedure

Step 1 Calculate the SOFR rate for each payment date and compute the floating cash flow at each date.

Step 2 Subtract the fixed cash flow from the floating cash flow (or the other way around) at each date to get the net cash flow.

Step 3 Discount all net cash flows at the zero rate. Sum = swap value.

Valuation of Interest Rate Swaps

Pay fixed 3% p.a. semiannual, receive SOFR — $100m notional, 1.25 years remaining

Given

Remaining life: 1.25 years | Payments at: 0.25, 0.75, 1.25 years
Zero rates (continuous): 3m = 2.8%, 9m = 3.2%, 15m = 3.4%

SOFR: for the next payment in 3 months = 2.9% (s.a.), 3–9m = 3.429% (s.a.), 9–15m = 3.734% (s.a.)

Time (yrs)	Fixed CF ($m)	Floating CF ($m)	Net CF ($m)	Discount Factor	PV of Net CF ($m)
0.25	−1.5000	+1.4500	−0.0500	0.9930	−0.0497
0.75	−1.5000	+1.7145	+0.2145	0.9763	+0.2094
1.25	−1.5000	+1.8672	+0.3672	0.9584	+0.3519
Total					**+0.5117**

Discount Factor = e^(-rt) where r is zero rate
Swap value = $0.5117 million (positive to the floating-rate payer)

Next Week

Guest lectures Adrian Dudek and Will Green from Optiver